

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	8.2 / 1.04082
Time	2026-07-08 02:00 UTC
Location	29.5831, -95.0155
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	6 / 621	39.12 to 100; mean 72.8144	88.61 at 2026-07-08 01:55Z; dt 5 min
asos	temperature	degC	6 / 621	23 to 36.1111; mean 29.008	24 at 2026-07-08 01:55Z; dt 5 min
asos	wind_direction	degree	6 / 715	0 to 340; mean 134	20 at 2026-07-08 01:55Z; dt 5 min
asos	wind_speed	m/s	6 / 749	0 to 10.8033; mean 2.3002	5.65888 at 2026-07-08 01:55Z; dt 5 min
noaa_gfs	gh_500	m	6 / 72	5896.72 to 5924.75; mean 5910.76	5912.45 at 2026-07-08 00:00Z; dt 120 min
noaa_gfs	pbl_height	m	6 / 72	28.4394 to 2153.73; mean 685.834	814.326 at 2026-07-08 00:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	6 / 72	39.1305 to 51.7657; mean 46.5519	47.2795 at 2026-07-08 00:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	6 / 72	1011.03 to 1017.68; mean 1014.98	1015.28 at 2026-07-08 00:00Z; dt 120 min
noaa_gfs	t_850	degC	6 / 72	17.233 to 20.4926; mean 19.0311	19.5621 at 2026-07-08 00:00Z; dt 120 min
noaa_gfs	u_10m	m/s	6 / 72	-2.38144 to 1.93714; mean 0.1802	-0.888887 at 2026-07-08 00:00Z; dt 120 min
noaa_gfs	v_10m	m/s	6 / 72	-0.854414 to 4.60966; mean 2.1245	4.0894 at 2026-07-08 00:00Z; dt 120 min
openaq	ozone	ppm	15 / 673	-0.004 to 0.069; mean 0.0172	0.017 at 2026-07-08 02:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 146	8 to 75; mean 25.6301	15 at 2026-07-08 02:00Z; dt 0 min
openaq	pm25	ug/m3	67 / 2374	0 to 50.7; mean 5.3055	2.2 at 2026-07-08 02:00Z; dt 0 min
openweather	cloud_cover	percent	3 / 215	0 to 100; mean 23.8372	33 at 2026-07-08 01:58Z; dt 1.1 min
openweather	humidity	percent	3 / 215	38 to 95; mean 76.3395	74 at 2026-07-08 01:58Z; dt 1.1 min
openweather	precipitation	mm/h	3 / 215	0 to 5.62; mean 0.0695	5.62 at 2026-07-08 01:58Z; dt 1.1 min
openweather	pressure	hPa	3 / 215	1012 to 1019; mean 1016.35	1017 at 2026-07-08 01:58Z; dt 1.1 min
openweather	temperature	degC	3 / 215	24.19 to 35.68; mean 29.6074	27.82 at 2026-07-08 01:58Z; dt 1.1 min
openweather	wind_direction	degree	3 / 215	0 to 359; mean 199.651	170 at 2026-07-08 01:58Z; dt 1.1 min
openweather	wind_speed	m/s	3 / 215	0.36 to 8; mean 2.3713	4.09 at 2026-07-08 01:58Z; dt 1.1 min
purpleair	pm25	ug/m3	60 / 4138	0 to 146.4; mean 7.3667	6.9 at 2026-07-08 02:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0243723 to 0.0253077; mean 0.0249	0.0249873 at 2026-07-07 18:57Z; dt 422.4 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_hcho_column	mol/m^2	8 / 8	0.000174097 to 0.000235341; mean 0.0002	0.000202795 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_hcho_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	3.63106e-05 to 4.30727e-05; mean 0	4.10314e-05 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_so2_column	mol/m^2	8 / 8	-3.49304e-05 to 4.72775e-05; mean 0	2.51087e-05 at 2026-07-07 20:37Z; dt 322.4 min
sentinel5p	s5p_so2_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 20:37Z; dt 322.4 min
tceq	co	ppm	3 / 162	0 to 1; mean 0.2395	0.2 at 2026-07-08 02:00Z; dt 0 min
tceq	no2	ppb	11 / 579	-0.1 to 35.8; mean 6.3107	8.2 at 2026-07-08 02:00Z; dt 0 min
tceq	so2	ppb	3 / 154	-0.5 to 6.8; mean -0.0487	0.1 at 2026-07-08 02:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-06 12Z 62.5, 18Z 55.13, 07-07 00Z 76.52, 06Z 88.01, 12Z 69.83, 18Z 53.58, 07-08 00Z 78.5, 06Z 90.93, 12Z 69.98, 18Z 58.27, 07-09 00Z 76.88, 06Z 88.36, 12Z 82.39
asos	temperature	07-06 12Z 31.71, 18Z 32.77, 07-07 00Z 28.48, 06Z 26.02, 12Z 29.96, 18Z 32.88, 07-08 00Z 26.09, 06Z 24.77, 12Z 30.39, 18Z 31.92, 07-09 00Z 28.28, 06Z 25.73, 12Z 27.88
asos	wind_direction	07-06 12Z 200, 18Z 168.9, 07-07 00Z 129.8, 06Z 70.98, 12Z 128.9, 18Z 164.8, 07-08 00Z 116.1, 06Z 23.62, 12Z 182.7, 18Z 169.8, 07-09 00Z 146.4, 06Z 118.3, 12Z 143.3
asos	wind_speed	07-06 12Z 2.268, 18Z 4.084, 07-07 00Z 2.151, 06Z 0.844, 12Z 1.479, 18Z 3.523, 07-08 00Z 2.864, 06Z 0.3568, 12Z 2.121, 18Z 4.013, 07-09 00Z 2.643, 06Z 1.444, 12Z 1.617
noaa_gfs	gh_500	07-06 18Z 5903, 07-07 00Z 5899, 06Z 5905, 12Z 5899, 18Z 5914, 07-08 00Z 5912, 06Z 5918, 12Z 5912, 18Z 5924, 07-09 00Z 5914, 06Z 5919, 12Z 5910
noaa_gfs	pbl_height	07-06 18Z 1742, 07-07 00Z 788.1, 06Z 171.2, 12Z 77.48, 18Z 1802, 07-08 00Z 824.4, 06Z 201.3, 12Z 100.2, 18Z 1948, 07-09 00Z 214.1, 06Z 247.5, 12Z 113.4
noaa_gfs	precipitable_water	07-06 18Z 43.91, 07-07 00Z 47.6, 06Z 45.54, 12Z 46.61, 18Z 49.19, 07-08 00Z 49.19, 06Z 46.64, 12Z 46.81, 18Z 48.62, 07-09 00Z 45.51, 06Z 44.36, 12Z 44.64
noaa_gfs	surface_pressure	07-06 18Z 1014, 07-07 00Z 1012, 06Z 1014, 12Z 1015, 18Z 1016, 07-08 00Z 1014, 06Z 1016, 12Z 1017, 18Z 1016, 07-09 00Z 1014, 06Z 1015, 12Z 1015
noaa_gfs	t_850	07-06 18Z 18.1, 07-07 00Z 19.9, 06Z 19.74, 12Z 18.83, 18Z 17.67, 07-08 00Z 19.17, 06Z 19.63, 12Z 19.03, 18Z 18.03, 07-09 00Z 19.89, 06Z 19.42, 12Z 18.96
noaa_gfs	u_10m	07-06 18Z 0.2464, 07-07 00Z -1.198, 06Z 0.3297, 12Z 1.134, 18Z -0.8133, 07-08 00Z -1.616, 06Z 0.4156, 12Z 1.132, 18Z 0.4076, 07-09 00Z -0.07952, 06Z 1.055, 12Z 1.149

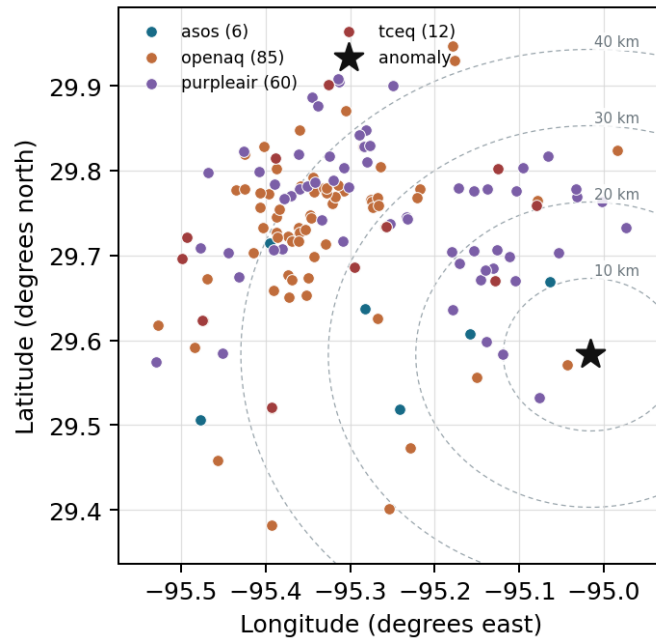
Source	Metric	Values
noaa_gfs	v_10m	07-06 18Z 1.711, 07-07 00Z 4.026, 06Z 2.07, 12Z 0.8139, 18Z 1.497, 07-08 00Z 4.036, 06Z 2.383, 12Z 0.8573, 18Z 0.8856, 07-09 00Z 3.191, 06Z 2.597, 12Z 1.426
openaq	ozone	07-06 12Z 0.0169, 18Z 0.03379, 07-07 00Z 0.01325, 06Z 0.004926, 12Z 0.01279, 18Z 0.02943, 07-08 00Z 0.02414, 06Z 0.004, 12Z 0.004111, 18Z 0.035, 07-09 12Z 0.009385
openaq	pm10	07-06 12Z 32.42, 18Z 27.5, 07-07 00Z 15.89, 06Z 18.28, 12Z 33.17, 18Z 29.88, 07-08 00Z 21.94, 06Z 24.39, 12Z 36, 18Z 23, 07-09 12Z 32.5
openaq	pm25	07-06 12Z 5.774, 18Z 5.265, 07-07 00Z 4.241, 06Z 5.95, 12Z 6.585, 18Z 7.067, 07-08 00Z 4.338, 06Z 6.606, 12Z 5.775, 18Z 3.028, 07-09 00Z 3.561, 06Z 4.493, 12Z 5.608
openweather	cloud_cover	07-06 12Z 7.455, 18Z 24.61, 07-07 00Z 5.842, 06Z 2.444, 12Z 19.44, 18Z 50.22, 07-08 00Z 36.28, 06Z 4.611, 12Z 3.765, 18Z 40.47, 07-09 00Z 70.16, 06Z 18.83, 12Z 5.167
openweather	humidity	07-06 12Z 70.91, 18Z 61.67, 07-07 00Z 76.58, 06Z 87.67, 12Z 76.33, 18Z 60.94, 07-08 00Z 77.89, 06Z 91.28, 12Z 78.35, 18Z 62.71, 07-09 00Z 75.95, 06Z 88.67, 12Z 89.5
openweather	precipitation	07-06 12Z 0, 18Z 0.008889, 07-07 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-08 00Z 0.7911, 06Z 0, 12Z 0, 18Z 0.03235, 07-09 00Z 0, 06Z 0, 12Z 0
openweather	pressure	07-06 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016, 12Z 1018, 18Z 1016, 07-08 00Z 1016, 06Z 1017, 12Z 1019, 18Z 1017, 07-09 00Z 1016, 06Z 1016, 12Z 1017
openweather	temperature	07-06 12Z 31.67, 18Z 34.19, 07-07 00Z 29.43, 06Z 26.47, 12Z 30.28, 18Z 34.05, 07-08 00Z 27.8, 06Z 24.8, 12Z 30.05, 18Z 33.25, 07-09 00Z 29.06, 06Z 26.26, 12Z 26.79
openweather	wind_direction	07-06 12Z 250.8, 18Z 180.4, 07-07 00Z 190.4, 06Z 212.7, 12Z 206.8, 18Z 186.3, 07-08 00Z 169.6, 06Z 201.8, 12Z 239.4, 18Z 192.8, 07-09 00Z 177.8, 06Z 205.6, 12Z 214.2
openweather	wind_speed	07-06 12Z 2.136, 18Z 2.329, 07-07 00Z 2.734, 06Z 1.768, 12Z 1.452, 18Z 2.861, 07-08 00Z 2.998, 06Z 2.081, 12Z 2.04, 18Z 2.951, 07-09 00Z 3.045, 06Z 2.073, 12Z 1.925
purpleair	pm25	07-06 12Z 8.147, 18Z 6.49, 07-07 00Z 6.121, 06Z 7.782, 12Z 8.078, 18Z 8.724, 07-08 00Z 7.736, 06Z 9.811, 12Z 7.884, 18Z 5.593, 07-09 00Z 5.09, 06Z 6.826, 12Z 8.346
sentinel5p	s5p_aer_ai_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_ch4_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_co_column	07-06 18Z 0.02528, 07-07 18Z 0.02495, 07-08 18Z 0.02437
sentinel5p	s5p_co_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_hcho_column	07-06 18Z 0.0002254, 07-07 18Z 0.0002152, 07-08 18Z 0.000185
sentinel5p	s5p_hcho_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_no2_column	07-06 18Z 4.086e-05, 07-07 18Z 3.867e-05, 07-08 18Z 4.307e-05
sentinel5p	s5p_no2_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_o3_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	s5p_so2_column	07-06 18Z 4.559e-05, 07-07 18Z 2.628e-06, 07-08 18Z -2.084e-05
sentinel5p	s5p_so2_granule_available	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
tceq	co	07-06 12Z 0.2364, 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.25, 12Z 0.2167, 18Z 0.2706, 07-08 00Z 0.2, 06Z 0.2444, 12Z 0.25, 07-09 06Z 0.2222, 12Z 0.3667
tceq	no2	07-06 12Z 4.832, 18Z 4.021, 07-07 00Z 3.418, 06Z 5.972, 12Z 5.721, 18Z 7.052, 07-08 00Z 6.258, 06Z 10.67, 12Z 7.524, 18Z 1.3, 07-09 06Z 5.853, 12Z 7.048
tceq	so2	07-06 12Z 0.2545, 18Z 0.075, 07-07 00Z -0.23, 06Z -0.2687, 12Z 0.3333, 18Z -0.05294, 07-08 00Z -0.2, 06Z -0.1867, 12Z -0.03889, 07-09 06Z -0.1556, 12Z -0.1

Per-site averages

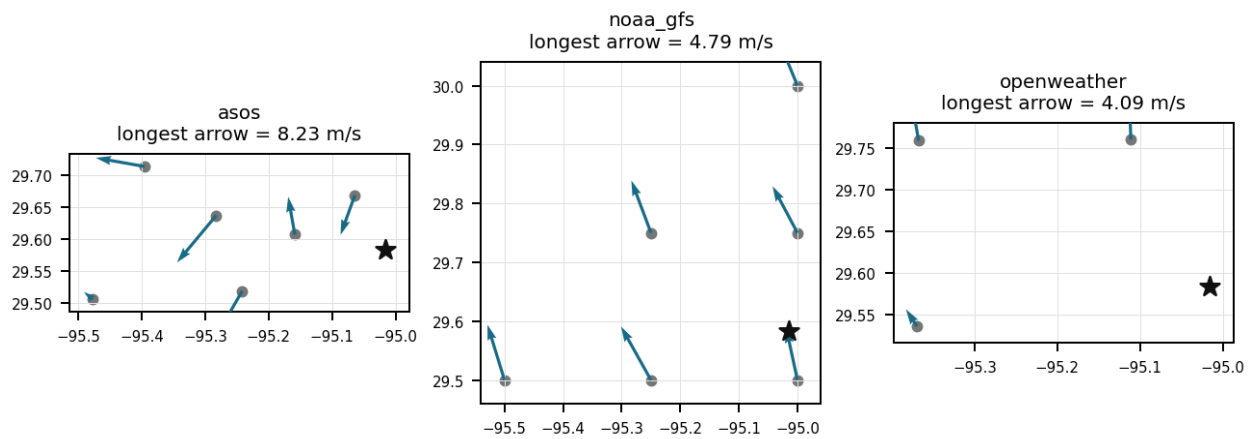
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (10.667 km) 75.04, EFD (14.102 km) 68.68, LVJ (23.01 km) 71.92, HOU (26.499 km) 72.76, MCJ (39.455 km) 72.42, AXH (45.443 km) 73.08
asos	temperature	T41 (10.667 km) 28.9, EFD (14.102 km) 30.37, LVJ (23.01 km) 29.18, HOU (26.499 km) 29.36, MCJ (39.455 km) 29.02, AXH (45.443 km) 28.32
asos	wind_direction	T41 (10.667 km) 164.5, EFD (14.102 km) 149.4, LVJ (23.01 km) 116.1, HOU (26.499 km) 165.6, MCJ (39.455 km) 140, AXH (45.443 km) 79.5
asos	wind_speed	T41 (10.667 km) 2.658, EFD (14.102 km) 3.223, LVJ (23.01 km) 1.808, HOU (26.499 km) 2.881, MCJ (39.455 km) 2.883, AXH (45.443 km) 0.9216
noaa_gfs	gh_500	gfs:29.50,-95.00 (9.357 km) 5911, gfs:29.75,-95.00 (18.624 km) 5911, gfs:29.50,-95.25 (24.49 km) 5911, gfs:29.75,-95.25 (29.288 km) 5911, gfs:30.00,-95.00 (46.387 km) 5910, gfs:29.50,-95.50 (47.768 km) 5911
noaa_gfs	pbl_height	gfs:29.50,-95.00 (9.357 km) 639.2, gfs:29.75,-95.00 (18.624 km) 755.9, gfs:29.50,-95.25 (24.49 km) 632.2, gfs:29.75,-95.25 (29.288 km) 872.2, gfs:30.00,-95.00 (46.387 km) 530.9, gfs:29.50,-95.50 (47.768 km) 684.5
noaa_gfs	precipitable_water	gfs:29.50,-95.00 (9.357 km) 46.95, gfs:29.75,-95.00 (18.624 km) 47.24, gfs:29.50,-95.25 (24.49 km) 46.16, gfs:29.75,-95.25 (29.288 km) 46.69, gfs:30.00,-95.00 (46.387 km) 47.03, gfs:29.50,-95.50 (47.768 km) 45.24
noaa_gfs	surface_pressure	gfs:29.50,-95.00 (9.357 km) 1016, gfs:29.75,-95.00 (18.624 km) 1016, gfs:29.50,-95.25 (24.49 km) 1015, gfs:29.75,-95.25 (29.288 km) 1015, gfs:30.00,-95.00 (46.387 km) 1014, gfs:29.50,-95.50 (47.768 km) 1014
noaa_gfs	t_850	gfs:29.50,-95.00 (9.357 km) 19.1, gfs:29.75,-95.00 (18.624 km) 19.09, gfs:29.50,-95.25 (24.49 km) 19.07, gfs:29.75,-95.25 (29.288 km) 19.02, gfs:30.00,-95.00 (46.387 km) 18.89, gfs:29.50,-95.50 (47.768 km) 19.02
noaa_gfs	u_10m	gfs:29.50,-95.00 (9.357 km) 0.2049, gfs:29.75,-95.00 (18.624 km) 0.05326, gfs:29.50,-95.25 (24.49 km) 0.1358, gfs:29.75,-95.25 (29.288 km) 0.3008, gfs:30.00,-95.00 (46.387 km) 0.2799, gfs:29.50,-95.50 (47.768 km) 0.1066
noaa_gfs	v_10m	gfs:29.50,-95.00 (9.357 km) 2.231, gfs:29.75,-95.00 (18.624 km) 2.076, gfs:29.50,-95.25 (24.49 km) 1.998, gfs:29.75,-95.25 (29.288 km) 2.263, gfs:30.00,-95.00 (46.387 km) 2.102, gfs:29.50,-95.50 (47.768 km) 2.077
openaq	ozone	5077791 (0.008 km) 0.01416, 332 (14.586 km) 0.02047, 2800 (21.051 km) 0.0186, 320 (24.786 km) 0.0147, 3214 (26.617 km) 0.01914, 339 (26.892 km) 0.02263, 2039 (28.563 km) 0.01665, 3378 (28.774 km) 0.01527, +7 more
openaq	pm10	271 (14.586 km) 16.8, 13380082 (28.774 km) 32.64, 15852419 (35.786 km) 27.92
openaq	pm25	14404954 (2.969 km) 3.349, 13955815 (13.307 km) 5.607, 268 (14.586 km) 6.928, 11638043 (23.95 km) 2.591, 14157975 (24.789 km) 7.538, 3379 (28.774 km) 8.363, 15539682 (28.776 km) 6.514, 8967123 (28.83 km) 7.325, +59 more
openweather	cloud_cover	grid:east (21.774 km) 23.17, grid:south (34.665 km) 23.08, grid:center (39.242 km) 25.28
openweather	humidity	grid:east (21.774 km) 78.81, grid:south (34.665 km) 74.22, grid:center (39.242 km) 75.99
openweather	precipitation	grid:east (21.774 km) 0.1236, grid:south (34.665 km) 0.001944, grid:center (39.242 km) 0.08324
openweather	pressure	grid:east (21.774 km) 1016, grid:south (34.665 km) 1016, grid:center (39.242 km) 1016
openweather	temperature	grid:east (21.774 km) 29.55, grid:south (34.665 km) 29.5, grid:center (39.242 km) 29.77
openweather	wind_direction	grid:east (21.774 km) 194.2, grid:south (34.665 km) 201.6, grid:center (39.242 km) 203.2
openweather	wind_speed	grid:east (21.774 km) 2.874, grid:south (34.665 km) 2.166, grid:center (39.242 km) 2.07
purpleair	pm25	2386 (8.161 km) 0.1, 3298 (10.11 km) 6.422, 247283 (12.095 km) 5.975, 268165 (12.925 km) 7.404, 222815 (13.892 km) 5.413, 222593 (15.865 km) 6.875, 128007 (15.888 km) 1.464, 161689 (15.943 km) 5.56, +52 more
tceq	co	482011039 (14.566 km) 0.1891, 482011035 (28.771 km) 0.1216, 482011052 (44.211 km) 0.3964
tceq	no2	482011050 (0.0 km) 1.511, 482011039 (14.566 km) 4.651, 482011015 (20.507 km) 7.456, 482010026 (26.633 km) 7.915, 482011035 (28.771 km) 10.66, 482010416 (29.325 km) 4.904, 480391004 (37.123 km) 2.68, 482011052 (44.211 km) 14.04, +3 more
tceq	so2	482011039 (14.566 km) 0.04792, 482011035 (28.771 km) -0.05962, 482010051 (44.59 km) -0.1241

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

Industrial NO_x emissions from nearby petrochemical facilities along the Houston Ship Channel accumulated locally under weak surface winds, contributing to the NO₂ anomaly recorded by tceq.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 2 of 35

The TCEQ NO₂ anomaly is strongly consistent with a localized NO_x plume because the anomalous monitor at 29.583, -95.016 measured 8.2 ppb against an expected 1.04 ppb while several other nearby TCEQ NO₂ monitors had much higher typical means, indicating sharp spatial heterogeneity rather than a uniformly elevated regional field.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 3 of 35

A local fresh NO_x emission plume from nearby combustion sources such as roadway traffic, diesel equipment, marine activity, or industrial combustion is a plausible cause of the sharp NO₂ enhancement because the anomalous monitor value greatly exceeded that monitor's expected baseline while co-located regional indicators remained modest and ozone was only about 0.017 ppm at the same time, which is consistent with recently emitted NO titrating ozone and producing elevated near-source NO₂.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 4 of 35

Evening mobile source emissions from regional transportation corridors contributed ground-level NO₂ precursors that concentrated near the tceq monitoring location.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 5 of 35

The anomaly occurred on July 8th at around 2:00 AM CDT.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 6 of 35

A severe thunderstorm or tornado event in the region has stirred up dust and pollutants, leading to elevated air quality readings.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 7 of 35

The air quality anomaly is related to PM_{2.5} levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 8 of 35

Planetary boundary layer height decreased significantly during the evening transition on July 8, 2026, restricting atmospheric vertical mixing in the Houston region.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 9 of 35

The wind direction and speed in the area were normal on July 8th, with no unusual weather patterns or events that could have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 10 of 35

Nighttime boundary-layer stabilization and reduced vertical mixing likely amplified near-surface NO₂ at the Houston site because humidity was high near 89 percent, temperature was about 24 C, and forecast planetary boundary layer height dropped from about 814 m near 00Z toward about 201 m by 06Z, conditions that favor trapping of primary pollutants close to the surface.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 11 of 35

The anomalous pollutant was nitrogen dioxide measured by tceq at 2026-07-08T02:00:00+00:00 near 29.583, -95.016 in Houston.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 12 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 13 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 14 of 35

A severe NO2 concentration anomaly reaching 8.2 ppb was observed at coordinates (29.583, -95.016) on July 8, 2026, at 02:00 UTC.

Mark exactly one:☐ V☐ I☐ U**Note:**

Claim 15 of 35

Nighttime meteorology likely enhanced accumulation of NO₂ near the surface because humidity was high near the anomaly time, temperatures were relatively cool, and modeled planetary boundary layer height dropped from about 824 m near 00Z to about 201 m by 06Z and near 100 m by 12Z, which is consistent with increasing nocturnal stability during the event window.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 16 of 35

Meteorology near 2026-07-08T02:00:00+00:00 favored transport and partial trapping of NO_x because surface winds were from the north to northeast near the site while the overnight boundary layer became shallow later in the night, a pattern consistent with plume impact from the Houston Ship Channel or nearby industrial sources.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 17 of 35

Transport from the Houston Ship Channel or adjacent industrial corridor into the monitor is a plausible contributor because the anomaly site lies in eastern Houston and local winds near the event shifted to northerly to northeasterly directions, with ASOS showing about 20 degrees at 01:55 UTC and GFS indicating flow with negative u and positive v around 00Z, a pattern capable of carrying emissions from industrial areas toward the site.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 18 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 19 of 35

The tceq NO2 measurement at the anomaly time and location was 8.2 ppb, while the expected value was 1.0408163265306123 ppb.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 20 of 35

The tceq NO2 anomaly at 2026-07-08T02:00:00+00:00 had a z-score of 11.19888831390569 and was classified as severe.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 21 of 35

NOAA GFS planetary boundary layer height data and ASOS wind observations confirm nocturnal atmospheric stabilization and low wind speeds around 02:00 UTC on July 8, 2026, which restricted vertical pollutant dispersion near the surface.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 22 of 35

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to the formation of ground-level ozone and particulate matter.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 23 of 35

The measured NO₂ concentration of 8.2 ppb significantly exceeded the expected baseline level of approximately 1.04 ppb, corresponding to a z-score of 11.2.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 24 of 35

Transport from the Houston Ship Channel or industrial corridor is plausible but not definitive because observed wind direction near the anomaly shifted toward northerly to northeasterly flow around 02Z–06Z while co-pollutants that often indicate a larger combustion or industrial episode, including CO, SO₂, PM_{2.5}, PM₁₀, and satellite CO and NO₂ columns, did not show corresponding strong regional enhancement.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 25 of 35

The TCEQ NO2 monitor at 29.583, -95.016 recorded 8.2 ppb at 2026-07-08T02:00:00+00:00, which was far above the monitor-specific expected value of about 1.04 ppb and is most consistent with a localized NOx enhancement rather than normal variability.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 26 of 35

Regional-scale combustion or smoke is not the leading explanation because ozone near the site was modest, PM2.5 and PM10 were not elevated, carbon monoxide remained low, and Sentinel-5P did not show an unusually large regional NO2 or CO signal in the nearest available overpass data.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 27 of 35

A sharp drop in planetary boundary layer height during the evening transition restricted vertical atmospheric mixing and trapped pollutants near the surface, as indicated by gfs data.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 28 of 35

Surface meteorological conditions recorded light surface winds and high relative humidity near the anomaly site during the night of July 8, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 29 of 35

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter and volatile organic compounds into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 30 of 35

Sentinel-5P satellite measurements detected an increased NO2 tropospheric column density over the Houston area on July 8, 2026, relative to previous days.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 31 of 35

TCEQ surface monitor network data show a regional increase in nighttime NO2 concentrations during the 06Z period of July 8, 2026, alongside stable low levels of carbon monoxide and sulfur dioxide.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 32 of 35

The PM2.5 levels are elevated at 6.9 ug/m3, which is higher than the mean of 7.3667 ug/m3.

Mark exactly one:

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Note:

Claim 33 of 35

A severe nitrogen dioxide concentration anomaly reaching 8.2 ppb was detected at the TCEQ monitoring station located at coordinates (29.583, -95.016) on July 8, 2026, at 02:00 UTC.

Mark exactly one:

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Note:

Claim 34 of 35

The air quality anomaly in Houston, Texas on July 8th is characterized by elevated levels of PM2.5.

Mark exactly one:

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Note:

Claim 35 of 35

The air quality anomaly in Houston, Texas on July 8th is also characterized by elevated levels of NO2.

Mark exactly one:

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Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
